

## Machine Tool Technology

### Chapter-04(Part-1)

#### 1. Introduction to Surface Texture Indication of Roughness Symbols

Complete surface symbol chart,  
Grade and Numbers

Tolerances- Unilateral and Bilateral

#### Surface Texture/Surface Finish/Surface Topography:

Repetitive or random deviations from the nominal surface (true flat surface) which form the pattern of the surface. Surface texture includes roughness, waviness, lay and flaws.

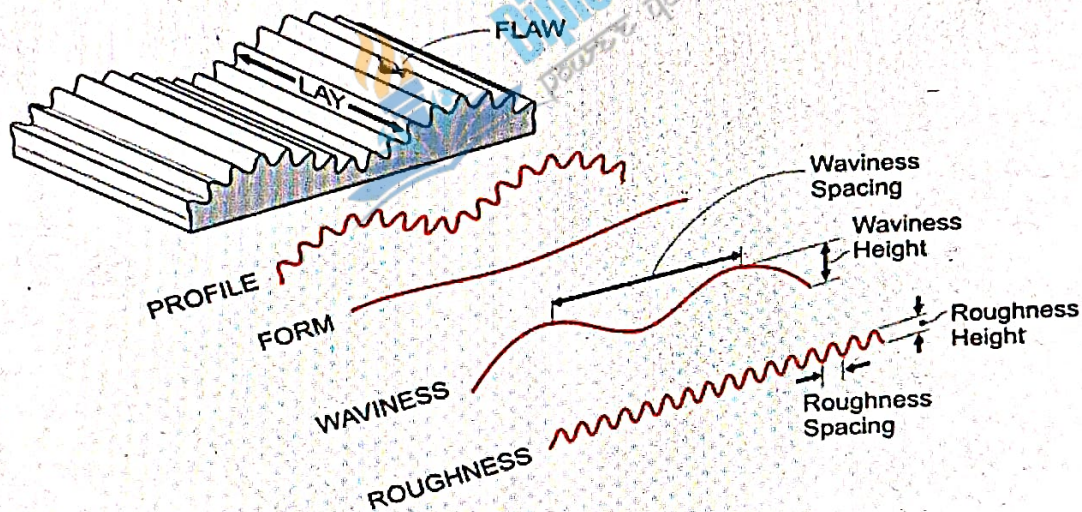
#### Surface Roughness/Roughness:

It is a component of surface texture.

It concerns all those irregularities which form surface relief and which are conventionally defined within the area where deviations of "form" and "waviness" are eliminated.

**Surface roughness** or roughness is defined as the irregularities which are inherent in the production process (e.g. cutting tool or abrasive grit).

Surface roughness is a calculation of the relative smoothness of a surface's profile.



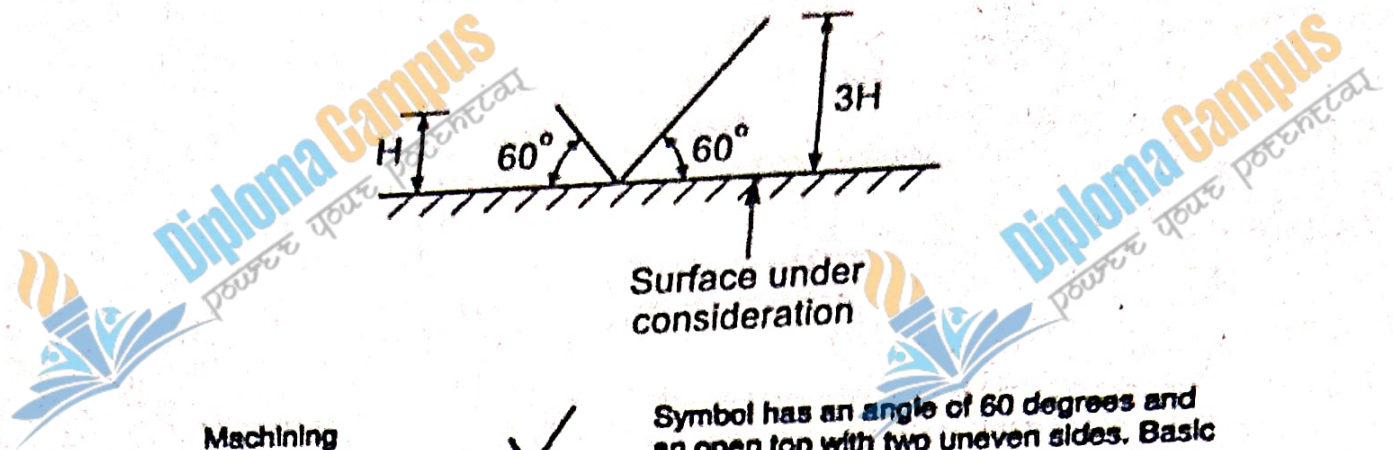
#### The factors affecting surface roughness

- Type of coolant used
- Cutting parameters such as feed, speed and depth of cut
- Type of machining
- Rigidity of the system consisting of machine tool, fixture, cutting tool and work
- Vibrations
- Material of tool and work piece.

Machine symbol



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Machining optional



Symbol has an angle of 60 degrees and an open top with two uneven sides. Basic shaped symbol.

Machine finish required



This symbol has a bar added to the top of the standard shaped symbol.

Removal of material not permitted



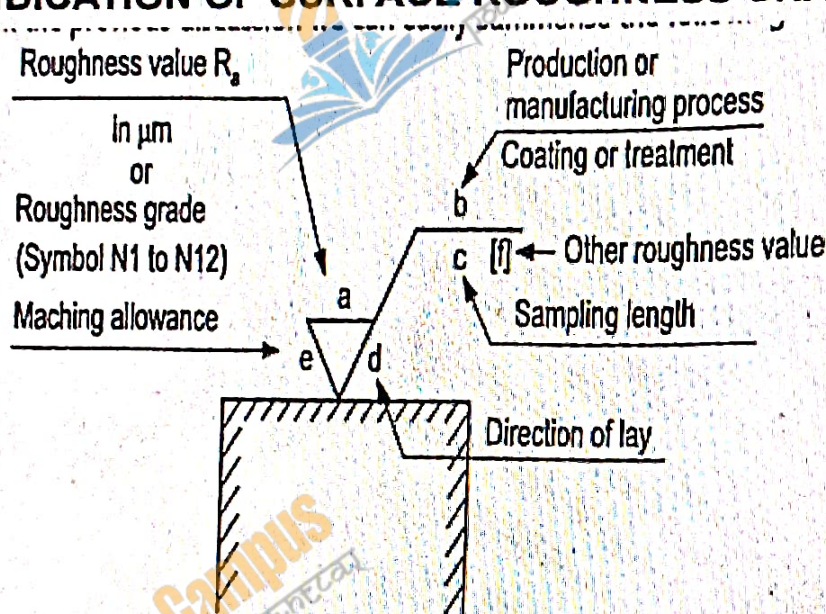
This symbol is a basic shaped symbol with a circle added in the groove.

Special surface feature required

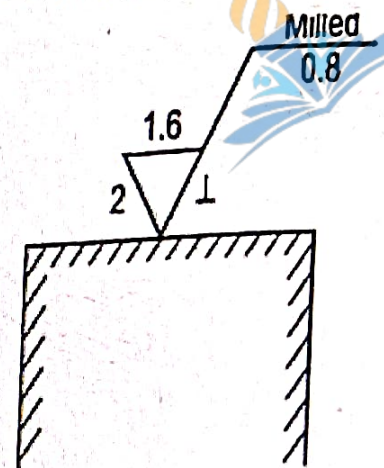


Basic symbol with a bar added to the top of the right hand line.

## INDICATION OF SURFACE ROUGHNESS CHARACTERISTICS



(a) Indication of machining symbol with all parameters

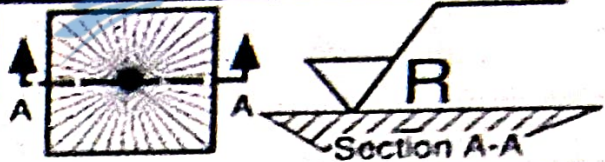
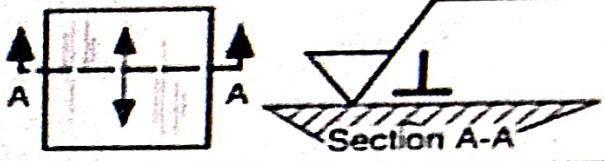
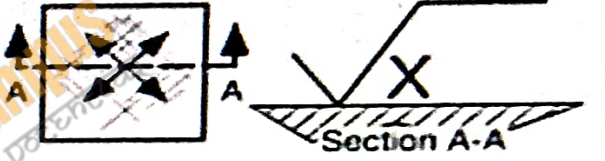
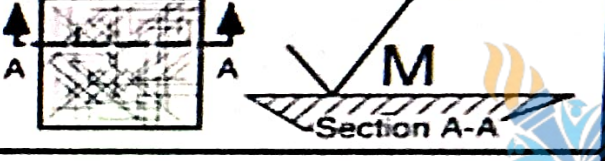
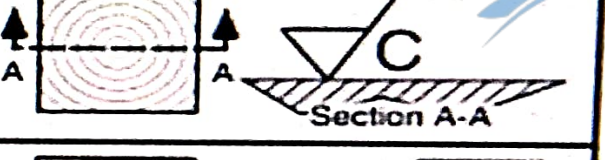
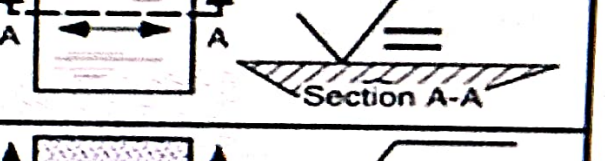
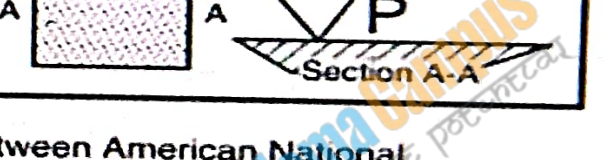


(b) Example

Surface roughness symbols and abbreviations

| <i>Roughness values<br/><math>R_a \mu m</math></i> | <i>Roughness<br/>grade number</i> | <i>Roughness<br/>grade symbol</i> |
|----------------------------------------------------|-----------------------------------|-----------------------------------|
| 50                                                 | N12                               | ~                                 |
| 25                                                 | N11                               | ▽                                 |
| 12.5                                               | N10                               |                                   |
| 6.3                                                | N9                                | ▽▽                                |
| 3.2                                                | N8                                |                                   |
| 1.6                                                | N7                                |                                   |
| 0.8                                                | N6                                | ▽▽▽                               |
| 0.4                                                | N5                                |                                   |
| 0.2                                                | N4                                |                                   |
| 0.1                                                | N3                                | ▽▽▽▽                              |
| 0.05                                               | N2                                |                                   |
| 0.025                                              | N1                                |                                   |

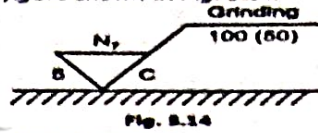
# Surface Texture - Lay Symbols Definitions (a)

| Symbol | Meaning                                                                                                                     | Direction of tool marks                                                              |
|--------|-----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| R      | Radial lay orientation relative to the center of the surface displaying the surface texture symbol.                         |    |
| ⊥      | Perpendicular orientation relative to the surface in the view displaying the surface texture symbol.                        |    |
| X      | Angular lay orientation in both directions relative to the surface in the view displaying the surface texture symbol.       |   |
| M      | Multidirectional lay orientation relative to the surface in the view displaying the surface texture symbol.                 |  |
| C      | Circular lay orientation relative to the center of the surface displaying the surface texture symbol.                       |  |
| =      | Parallel lay orientation relative to the surface in the view displaying the surface texture symbol.                         |  |
| P      | Particulate, non-directional, or protuberant lay orientation relative to the surface displaying the surface texture symbol. |  |

(a) The symbols presented are consistent between American National Standards Committee B46 (Surface Texture) and ISO 1302, "Methods of Indicating Surface Texture on Drawings."

PROBLEMS:-

**Problem 5.5 :** State the meaning of the figure shown in Fig. 5.14.



**Solution :** Grinding – Manufacturing process  
 $N_7$  – Roughness grade with roughness value 1-5  $\mu\text{m}$   
 $S$  – Machining allowance  
 $e$  – Direction of lay (Circular)

**Problem 5.6 :** Represent with a neat sketch the surface roughness with following parameters. (i) Roughness value is 12.5  $\mu\text{m}$ . (ii) Parallel direction of lay. (iii) Machining by milling. (iv) Sampling length of 0.8 mm.

**Solution :**

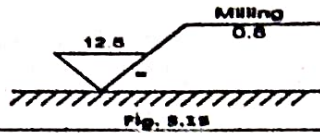


Fig. 5.15

**Problem 5.7 :** Draw suitable machining symbol and indicate following machining parameters on it. Machining method is milling, surface roughness value is 12  $\mu\text{m}$ , machining allowance is 5 and direction of lay is circular.

**Solution :**

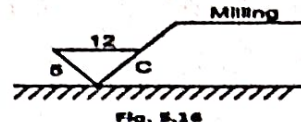


Fig. 5.16

**Problem 5.8 :** State the meaning of the symbol shown in Fig. 5.17.

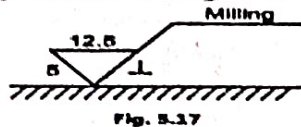


Fig. 5.17

**Solution :** Milling – Manufacturing process.  
 12.5 – Roughness value ( $R_a$ ) in  $\mu\text{m}$  (micron) or roughness grade.  
 $S$  – Machining allowance in mm  
 $e$  – Direction of lay.

**Problem 5.5 :** State the meaning of the figure shown in Fig. 5.14.



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 $N_7$  – Roughness grade with roughness value 1-5  $\mu\text{m}$   
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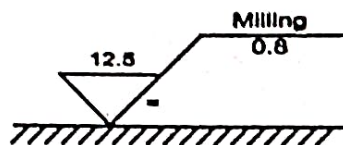


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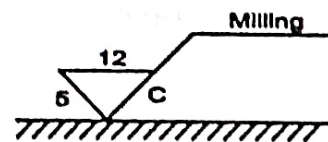


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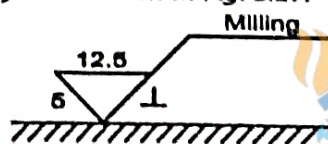


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## TOLERANCE

Tolerance is the permissible variation in the dimensions of a component (hole or shaft). Tolerances can be divided into Two parts, as follows.

1. Unilateral Tolerance
2. Bilateral Tolerance

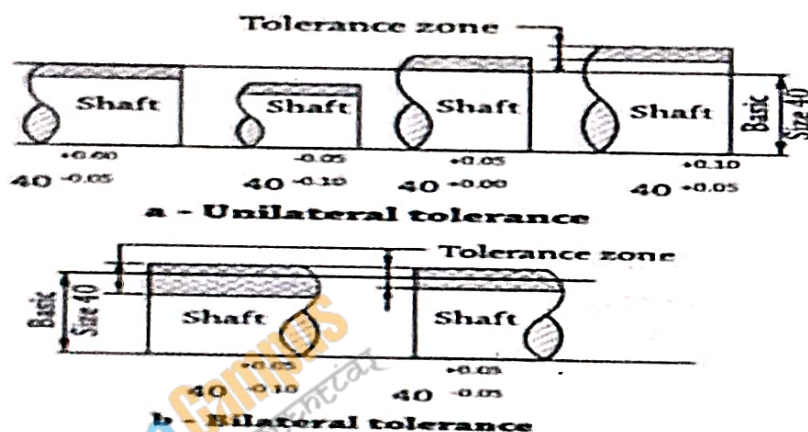
### 1. Unilateral Tolerance

In this type of system, the dimension of a part is allowed to vary only on one side of the basic

size of the work which means, the limits of tolerance on one side of the basic size of the work either above of the work or below of the work

### 2. Bilateral Tolerance

In this system, the dimension of the part is allowed to vary on both sides of the basic size.



Visual Watermark

## Machine Tool Technology Chapter - 4(Part-2)

### 2. Introduction to Lathe

#### Types of lathe

Capstone and Turret Lathe,  
Automatic Lathe and others

#### Specification of a Center Lathe

#### Cutting Fluids- Selection of cutting fluids

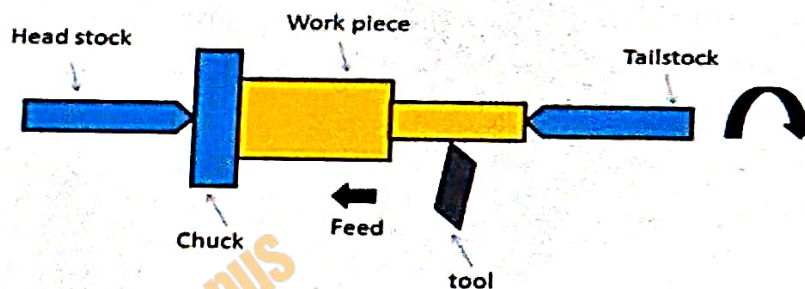
### 3. Explain Work holding Devices- Tool Holding Devices

#### What is a Lathe Machine?

A lathe machine is a machine tool that is used to remove unwanted material from a rotating workpiece in the form of chips.

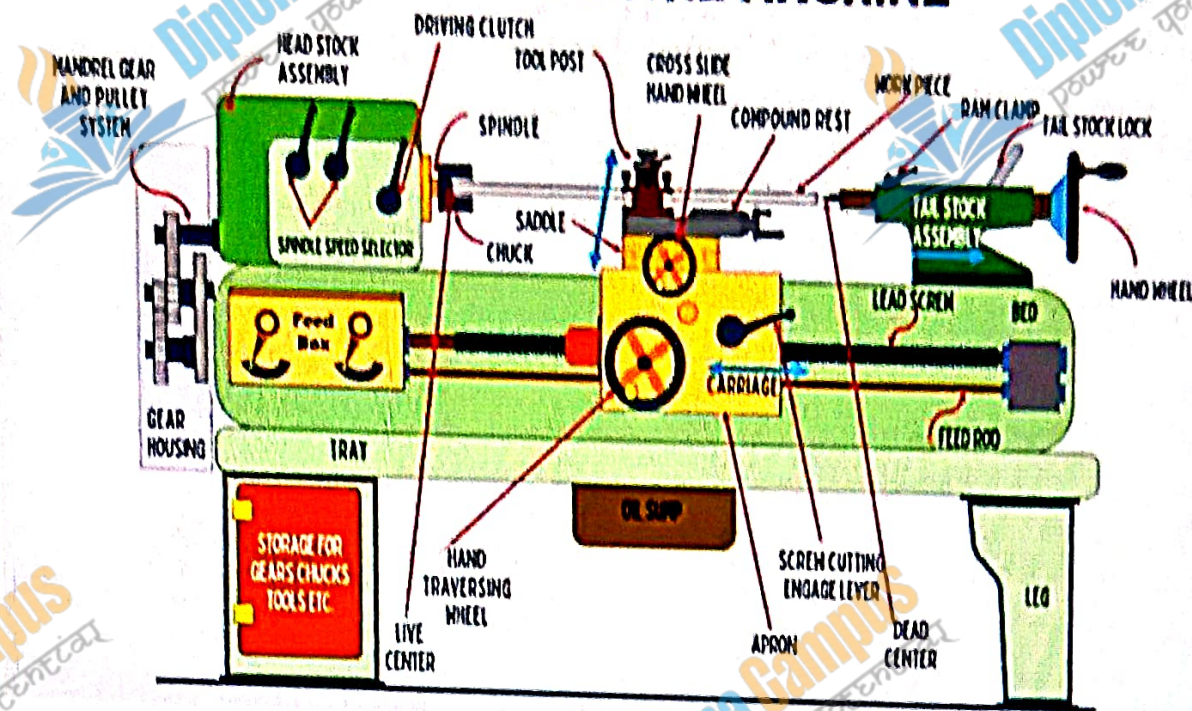
The machine is used to accomplish all fundamental operations, including drilling, sawing, tapping, and turning, among others, with the aid of various tools located in the work area. Hence, the lathe is also referred to as the "Mother of all machines."

**Working Principle:** The lathe is a machine tool which holds the workpiece between two rigid and strong supports called centers or in a chuck or face plate which revolves. The cutting tool is rigidly held and supported in a tool post which is fed against the revolving work. The normal cutting operations are performed with the cutting tool fed either parallel or at right angles to the axis of the work.



#### Lathe Machine Parts(Specification of centre Lathe)

# PARTS OF LATHE MACHINE



## Headstock

The headstock is a critical component of the lathe, located at the machine's left end. It houses the main spindle, gears, and often the motor that drives the spindle. The primary function of the headstock is to hold and rotate the workpiece. The speed at which the spindle turns can often be adjusted, providing control over the cutting speed.

## Tailstock

Function: Positioned opposite the headstock, the tailstock supports the other end of the workpiece.

Role: It's commonly used to hold tools like drills or reamers for operations such as drilling. The tailstock can be moved along the lathe bed and clamped in position, providing additional support for longer workpieces.

## Bed

Function: The bed of the lathe serves as a robust base that supports the major components of the machine, including the headstock, tailstock, and carriage.

Role: It ensures alignment and precision by holding these components in place. The bed must be extremely stable and rigid to prevent any movement or vibration during operation, which could affect the accuracy of the work.

## Carriage

Function: The carriage slides along the bed and is responsible for carrying the cutting tool.

Role: It facilitates precise lateral (side to side) and longitudinal (back and forth) movements for cutting.



Visual Watermark

### **Lead Screw**

**Function:** A long threaded shaft that runs parallel to the bed.

**Role:** The lead screw is crucial for precise threading operations and movement of the carriage. It works in conjunction with the carriage's feed mechanism, allowing for controlled and precise cuts, especially in threading operations.

### **Feed Rod**

**Function:** The feed rod, often running alongside the lead screw, also runs parallel to the bed.

**Role:** It powers the carriage movement for non-threading operations. This rod is essential for the consistent movement of the carriage during turning operations, ensuring smooth and even cuts.

### **Spindle**

**Function:** Located inside the headstock, the spindle is rotated by the motor.

**Role:** It holds and rotates the workpiece or chuck. The spindle is the primary rotating component on which the workpiece is mounted, and its speed and rotation can be controlled to suit various machining tasks.

### **Chuck**

**Function:** Attached to the spindle, the chuck grips and holds the workpiece in place.

**Role:** Chucks can be a three-jaw (self-centering) or four-jaw (independently adjustable) type, allowing for different shapes and sizes of workpieces to be securely held during the machining process.

### **Tool Post**

**Function:** Mounted on the carriage, the tool post holds the cutting tool.

**Role:** It allows the tool to be positioned and secured at various angles, providing the flexibility to perform a range of cutting operations.

### **Cross Slide**

**Function:** Mounted on the carriage, the cross slide moves perpendicular to the bed.

**Role:** This component provides movement towards or away from the workpiece, crucial for depth control in cutting operations.

### **Compound Rest**

**Function:** The compound rest sits atop the cross slide and allows for angled cuts and fine tool adjustments.

**Role:** It is used for precise cutting angles and finishing work, providing the ability to make precise angular cuts and adjustments for complex machining tasks.

### **Apron**



**Function:** The apron is part of the carriage that houses the control mechanisms.

**Role:** It contains gears, clutches, and levers used to control the movement of the carriage and the cross slide.

### TYPES OF LATHE

**1) Speed Lath Machine:** It consists of bed, tailstock, headstock & tool post installed on the adjustable slide. It has no carriage or feed box lead screw. The speed of the lathe machine is 1200 to 3600 rpm. It even consists of two or three spindle speed. The depth of the same is less. This type of lathe machine is employed in spinning, woodworking, polishing and centering. The high speed of the headstock spindle has named it a Speed lathe machine.

**2) Engine Lath:** It is operated by the pulley mechanism or gear mechanism. It is one of the most used lathe machines. It consists of a bed, tailstock, head etc. The head stock even provides multiple spindles speed. Basically, the headstock is stiff in structure.

This type of lathe machine is utilised in workshops and many other industries. It is capable of feeding the cutting tool in the cross as well as a longitudinal direction while being connected with to the lathe axis with the help of leadscrew and carriage feed. It consists of three belts: Geared type, belt driven and motor driven

**3) Bench lathe:** These lathe machines are used for very detailed work and are normally small in size. These are mounted on bench and consist of similar parts like speed and engine lathe machines. It can perform all the operations of speed and engine lathe machine but the only visible difference is of size.

**4) Toolroom Lathe machine:** This lathe machine has similar features as of engine lathe machine but the only variation is of its wide range of speed. The spindle speed of the same accelerates from very low to a high speed that is up to 2500 rpm. This machine consists of taper turning attachment, chuck, draw in collect attachment, pump for coolant, steady and follower rest etc.

This machine is used for working on dies gauges, tool, grinding and for many other purposes. It is costlier than other lathe machines.

**5) Capstan Lath and Turret Lath machines:** The capstan lathe and turret lathe machines are employed in production work and are considered as an extension of engine machine. It consists of hexagonal turret head and three tool post. Multiple tools can be fitted and fed into the work piece. This machine supports different operations and can produce different parts at a fast pace. It occupies more floor space and is used in large industries only.

**6) Special purpose lathe:** These lathe machines are meant for special purpose uses in the production industry. It can perform operations like grinding, drilling, boring, reaming etc.



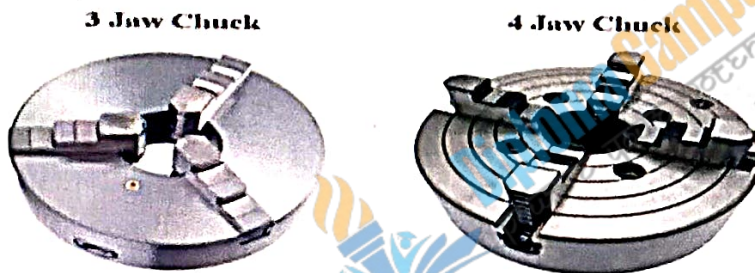
## Work-Holding devices used on Lathe

- 1 Chucks
- 2 Centres
- 3 Faceplates
- 4 Rests
- 5 Mandrels
- 6 Lathe Dogs

### 1.Chucks

A chuck is a device, which is used for holding and rotating the job of shorter length during machining. A chuck is usually equipped with three or four jaws and accordingly they are classified as three-jaw chucks and four-jaw chucks. These are shown in figure below.

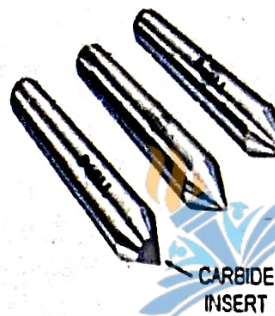
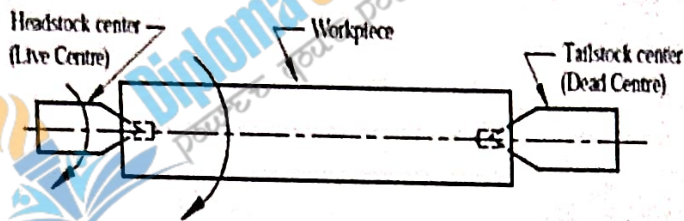
#### Different types of chucks



( I ) **Three-jaw chuck:** In a three-jaw chuck all the jaws are made to slide simultaneously by an equal amount within the slots provided on the body by rotating any one of the three pinions with the help of a handle. This is also known as a self-centering or a universal chuck. This chuck is suitable for holding round or hexagonal and other similar shaped workpieces. The gripping force of the 3-jaw chuck is less than that of the 4-jaw chuck.

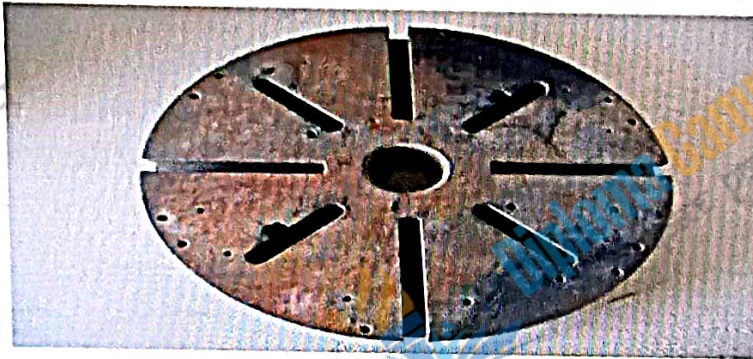
( II ) **Four-jaw chuck:** The 4-jaw chuck is a very versatile work holding device, able to hold cylindrical as well as odd-shaped parts for machining. In a four-jaw chuck, the jaws can be moved and adjusted independent of each other. The jaws are individually reversible to best match the shape of the workpiece. Hence, this is also known as independent jaw chuck. It can be also used for machining eccentrics by offsetting the workpiece. Its grip is more powerful than that of the 3-jaw chuck and gives better adjustment as you have control over each individual jaw.

### 2.Centres



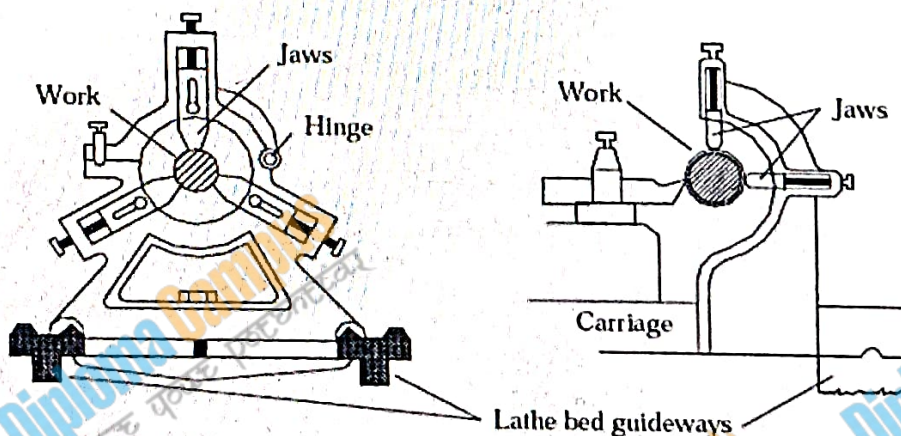
When the length of job large and if it cannot be conveniently held in a chuck, the workpiece can be held and rotated between the headstock centre (the live centre) and the tail stock centre (the dead centre). Turning between the centres is the classic way of machining the barstock with maximum accuracy. The work can be mounted and dismantled from the lathe quickly and without losing setup accuracy. The centre holes on workpiece mate with the points of the lathe centres.

### 3.Faceplates



Workpieces that cannot be accommodated in a chuck or between centres because they are asymmetrical or have a complex shape can be bolted to the faceplate either directly or indirectly via an angle plate. Bolts, clamps and dogs are used with the faceplate to hold the workpiece. The faceplate is directly mounted on the lathe spindle.

### 4.Resters



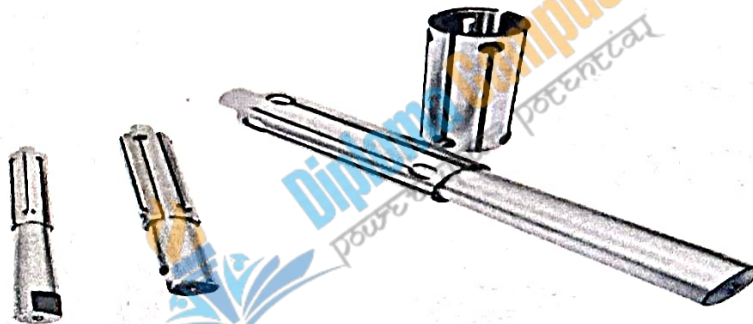
A rest is used for providing additional support to long workpiece when it is machined between centres or held in a chuck. If thin and long workpieces are not supported during machining, then there is a possibility that it may bend due to self-weight or due to the cutting forces exerted by the cutting tool on the workpiece. Two types of rests are in common use; these are the steady (or fixed) and follower (or traveling) rest.

( I ) **Steady rest:** This has three jaws, two on the lower base and one on the upper frame.

These jaws can be adjusted radially by rotating individual screws to accommodate workpieces of different diameters. These jaws act as bearing to the workpiece surface. The steady rest is clamped to the bed at the desired position, that is, where support is necessary. Depending on the length and weight of the workpiece more than one steady rest can also be used.

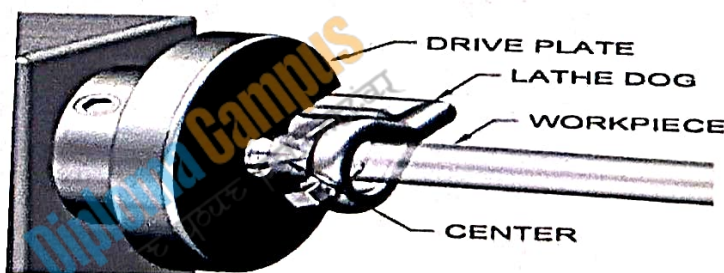
( II ) **Follower rest:** This rest is similar to steady rest but is bolted to the back end of the carriage. Unlike steady rest, which is stationary, the follower rest moves along with the carriage. The follower rest consists of a 'C' like casting having two adjustable jaws to support the workpiece.

## 5. Mandrels



A workpiece having a large hole in it like a pipe cannot be held firmly in a chuck, because of its reduced strength. In such cases, mandrels are used to hold the workpieces from inside. During machining, mandrel is held between centres and the workpiece is then machined similar to machining between centres.

## 6. Lathe Dogs



Lathe dogs are clamping devices used for holding workpiece or mandrels for turning between centres. The part to be held is inserted in the v-shaped hole and then firmly secured in position by means of the screw. These lathe dogs can hold the workpiece of only certain range of diameters. In case of large or small size, chucks or faceplates are or other work holding devices are used.

### **Cutting fluid**

Cutting fluids are liquids that are used in metal cutting operations to prevent the production of heat and reduce friction between the cutting tool and the workpiece. They play a crucial role in the machining process by cooling the tool and the workpiece and providing lubrication, which helps to reduce the cutting force and improve the surface finish.

Example - mineral oils, animal oils, vegetable oils, and fats etc.

### **Functions of Cutting Fluids**

The various functions of Cuttings Fluids are:

- Cooling the workpiece and cutting tool is the other primary function of cutting fluid.
- Reducing friction between the tool and workpiece.
- Facilitating chip evacuation from the cutting zone.
- Extending the longevity of cutting tools.
- Achieving smoother workpiece surfaces.
- Preventing workpiece and machine tool corrosion.
- Minimising material adhesion to the tool.
- Managing dust and fumes for safety.
- Expanding the range of cutting parameters.
- Enhancing the machining of various materials.

### **Types of Cutting Fluid**

There are four main types of cutting fluids:

- soluble cutting oil,
- neat cutting oil,
- synthetic coolants, and
- solid lubricants.

Soluble cutting oil is a type of cutting fluid that is soluble in water and is commonly used in large-scale operations.

Neat-cutting oil is a type of cutting fluid that is not soluble in water and is ideal for small-scale operations.

Synthetic coolants are cutting fluids made of synthetic materials and are used in applications where high performance is required.

Solid lubricants are cutting fluids in a solid form and are used in high-temperature applications.

### **Selection of Cutting Fluids**

- Primarily, it relies on the composition of both the workpiece material and the tools employed in the operation.





- The type of machining operation being conducted is another pivotal determinant in the selection of cutting fluids.
- Ultimately, the decision is influenced by the desired cooling effect necessary for the machining procedure.

